

**Mohammed Reabai J. 250701435-2025-CSE** ▾**M2****Started on** Monday, 24 August 2026, 8:35 PM**State** Finished**Completed on** Monday, 24 August 2026, 8:41 PM**Time taken** 5 mins 40 secs**Marks** 1.00/1.00**Grade** **10.00** out of 10.00 (**100%**)

**Question 1** | Correct | Mark 1.00 out of 1.00**Problem Statement**

Given an array of 1s and 0s this has all 1s first followed by all 0s. Aim is to find the number of 0s. Write a program using Divide and Conquer to Count the number of zeroes in the given array.

Input Format

First Line Contains Integer m – Size of array

Next m lines Contains m numbers – Elements of an array

Output Format

First Line Contains Integer – Number of zeroes present in the given array.

**Answer:** (penalty regime: 0 %)

```

1  #include <stdio.h>
2  int firstZero(int a[],int low,int high){
3  if(low>high){
4      return -1;
5  }
6  int mid=(low+high)/2;
7  if(a[mid]==0){
8      if(mid==0 || a[mid-1]==1){
9          return mid;
10     }
11     return firstZero(a,low,mid-1);
12 }
13 else{
14     return firstZero(a,mid+1,high);
15 }
16 }
17 int main(){
18     int m;
19     scanf("%d",&m);
20     int a[m];
21     for(int i=0;i<m;i++){
22         scanf("%d",&a[i]);
23     }
24     int pos=firstZero(a,0,m-1);
25     if(pos==-1){
26         printf("0");
27     }
28     else{
29         printf("%d",m-pos);
30     }
31     return 0;
32 }
```

|   | Input                      | Expected | Got |   |
|---|----------------------------|----------|-----|---|
| ✓ | 5<br>1<br>1<br>1<br>0<br>0 | 2        | 2   | ✓ |

|   | Input                                                                                             | Expected | Got |   |
|---|---------------------------------------------------------------------------------------------------|----------|-----|---|
| ✓ | 10<br>1<br>1<br>1<br>1<br>1<br>1<br>1<br>1<br>1<br>1<br>1<br>1                                    | 0        | 0   | ✓ |
| ✓ | 8<br>0<br>0<br>0<br>0<br>0<br>0<br>0<br>0<br>0                                                    | 8        | 8   | ✓ |
| ✓ | 17<br>1<br>1<br>1<br>1<br>1<br>1<br>1<br>1<br>1<br>1<br>1<br>1<br>1<br>1<br>1<br>1<br>1<br>0<br>0 | 2        | 2   | ✓ |

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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